INCH-POUND

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MILITARY SPECIFICATION

INITIATOR, PROPELLANT ACTUATED, M104 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M104 Propellant Actuated Initiator Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109 Quality Assurance Terms And Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

9252247	Carton, Packing Ammunition for, Initiator Propellant Actuated, M104
9252248	Box, Packing, Ammunition, for Initiator, Propellant Actuated, M104
9252249	Carton, Alternative, Packing Ammunition, for Initiator, Propellant Actuated, M104
10551447	Initiator, Propellant Actuated, M104, Metal Parts Assembly

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with see 4.4.

3.2 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

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3.4 Hydrostatic pressure. The assembled female and male bodies of the initiator shall withstand a hydrostatic pressure of 14,000 to 15,000 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.4.1).

3.5 Metal parts assembly. Parts shall be combined to produce an assembly conforming to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.6 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.7 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.4.

3.8 Workmanship. The requirements for workmanship shall be as specified on the applicable drawings, referenced specifications, and the following:

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice

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to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68° to 77°F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psia

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, and one set of unassembled parts without plating or final protective finish. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation, and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. The production lot size shall be as specified in the contract (see 6.2).

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4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Body, Male, Drawing 10551448, covering a detail of Drawing 10551447.

Characteristics	Defect	Method of Inspection
Critical		
1	Metal defective	Visual
Major		
101	Pitch diameter of small external thread, min	Gage
102	Major diameter of small external thread, min	Gage
103	Pitch diameter of large external thread, min	Gage
104	Major diameter of large external thread, min	Gage
105	Concentricity of flash hole with pitch diameter of small external thread	Gage
106	Concentricity of thread pilot of small external thread with pitch diameter of the small external thread	Gage
107	Concentricity of cavity with large external thread	Gage
108	Concentricity of pitch diameter of the large external thread with pitch diameter of the small external thread	Gage
109	Concentricity of body at large end with large external thread	Gage
110	Diameter of flash hole	Gage
111	Diameter across thread pilot	Gage
112	Diameter of shoulder	Gage
113	Diameter of large body	Gage
114	Diameter of cavity	Gage
115	Depth of cavity	Gage
116	Length to shoulder	Gage
117	Length of effective thread, min	Gage
118	Width of thread undercut	Gage
119	Thread damaged or not full	Gage
120	Concentricity of pitch diameter of large external thread with flashtube bore	Gage
121	Length of datum location	Gage

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122	Diameter of undercut of gasport		Gage
123	Protective finish	Visual	
124	Burr	Visual	
125	Foreign matter	Visual	
Minor			
201	Pitch diameter small external thread, max		Gage
202	Pitch diameter of large external thread, max		Gage
203	Length of thread relief to nose		Gage
204	Length from base to shoulder		Gage
205	Length of strip on shoulder	Gage	
206	Total length	Gage	
207	Length of thread pilot	Gage	
208	Width at nose chamfer	Gage	
209	Radius or chamfer missing or incorrect		Visual
210	Surface finish improper	Visual	
211	Chamfer on flashtube bore in counterbore incorrect		Gage

4.5.2.2 Body, Female, Drawing 10551449, covering a detail of Drawing 10551447.

Critical			
1	Metal defective		Visual
Major			
101	Pitch diameter internal cavity, max		Gage
102	Minor diameter internal cavity, max		Gage
103	Pitch diameter external thread, min		Gage
104	Major diameter external thread, min		Gage
105	Concentricity of external thread pilot with pitch diameter of external thread		Gage
106	Concentricity of flash hole with pitch diameter of external thread		Gage
107	Concentricity pitch diameter internal thread with pitch diameter of external thread		Gage
108	Concentricity of cavity with pitch diameter of internal thread		Gage
109	Concentricity of body with pitch diameter of internal thread		Gage
110	Diameter of flash hole	Gage	
111	Length of effective thread, min		Gage
112	Length of internal thread pilot		Gage
113	Width of undercut	Gage	
114	Length of datum location	Gage	
115	Concentricity of pitch diameter of internal thread with flashtube bore		Gage
116	Diameter of undercut of gasport		Gage
117	Thread damaged or not full	Visual	

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118	Protective finish inadequate or defective		Visual
119	Burr	Visual	
120	Foreign matter	Visual	
Minor			
201	Pitch diameter of internal cavity thread, max		Gage
202	Pitch diameter of external thread, max		Gage
203	Diameter across cut-out on body		Gage
204	Diameter across thread pilot		Gage
205	Diameter of cavity	Gage	
206	Diameter of body	Gage	
207	Depth of cavity	Gage	
208	Length of cut-out on body	Gage	
209	Length of thread pilot	Gage	
210	Length of port	Gage	
211	Total length	Gage	
212	Width at end of flash hole	Gage	
213	Surface finish improper	Visual	
214	Chamfer on flashtube bore in counterbore incorrect		Gage

4.5.2.3 Sealing, Ring, Drawing 10551451, covering a detail of Drawing 10551447.

Critical			
1	Metal defective		Visual
Major			
101	Diameter, inside, max	Gage	
102	Diameter, outside, min	Gage	
103	Protective finish inadequate or defective		Visual
104	Burr	Visual	
105	Foreign matter	Visual	
Minor			
201	Diameter, inside, min	Gage	
202	Diameter, outside, max	Gage	
203	Thickness	Gage	
204	Surface finish improper	Visual	

4.5.2.4 Sealing, Disc, Drawing 10551452, covering a detail of Drawing 10551447.

Critical			
1	Metal defective		Visual
Major			
101	Diameter, min	Gage	

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102	Protective finish	Visual
103	Burr	Visual
104	Foreign matter	Visual

Minor

201	Diameter, max	Gage
202	Thickness	Gage
203	Surface finish improper	Visual

4.5.2.5 Tube, Flash, Drawing 10551453, covering a detail of Drawing 10551447.

Critical

1	Verify that .116 diameter intersects with .062 holes, both ends	Visual/Gage
2	Metal defective	Visual

Major

101	Straightness	Gage
102	Diameter inside tube	Gage
103	Diameter outside tube	Gage
104	Diameter holes	Gage
105	Location of holes	Gage
106	Holes not equally spaced	Gage
107	Surface finish of tube	Visual
108	Protective finish improper or inadequate	Visual

Minor

201	Depth to bottom of tube	Gage
202	Total length	Gage
203	Diameter across flats at ends of flashtube, min	Gage
204	Chamfer missing or incorrect	Visual

4.5.2.6 Assembly, Drawing 10551447.

Critical

1	Metal defective	Visual
2	Any part missing	Visual
3	Improper assembly	Visual

Major

101	Protective finish inadequate or defective	Visual
102	Foreign matter	Visual

Minor

201	Length from undercut of female body to undercut	Gage

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	male body, max	
202	Adhesive missing from exterior surfaces	Visual

4.5.2.7 Carton, Packing Ammunition, Drawing 9252247.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.5.2.8 Box, Packing, Ammunition, Drawing 9252248.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of unit incorrect	Visual

4.5.2.9 Box, Fiberboard (Outer), Alternative, Packing Ammunition, Drawing 9252249.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units contained incorrect	Visual

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204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Ten units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. The assembled female and male bodies shall be subjected to the hydrostatic pressure test. Failure of any subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.4 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.4. Failure of any unit in the inspection sample to comply with 3.7 shall be cause for rejection of the assembly.

4.6 Test methods and procedures.

4.6.1 Hardness. Prior to application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to the application of protective finish, the male and female body assembly shall be mounted in the device, using a fixture for properly sealing all openings and subjected to the internal hydrostatic pressure to determine compliance to 3.4.

4.6.3 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.4 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and

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container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 9252247 and 9252248. Level B shall be in accordance with Drawings 9252247 and 9252249.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and revision letter of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. If first article inspection is required (see 3.1 and 6.3)

d. The responsible inspection activity (see 4.1)

e. Any special inspection requirements (see 4.3)

f. Level of packing and marking (see 5.1)

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of five metal parts assemblies, nine sets of unassembled parts, and one set of unassembled parts without plating or final protective finish. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate

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bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0E17)

Review activity:
Air Force - 70

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